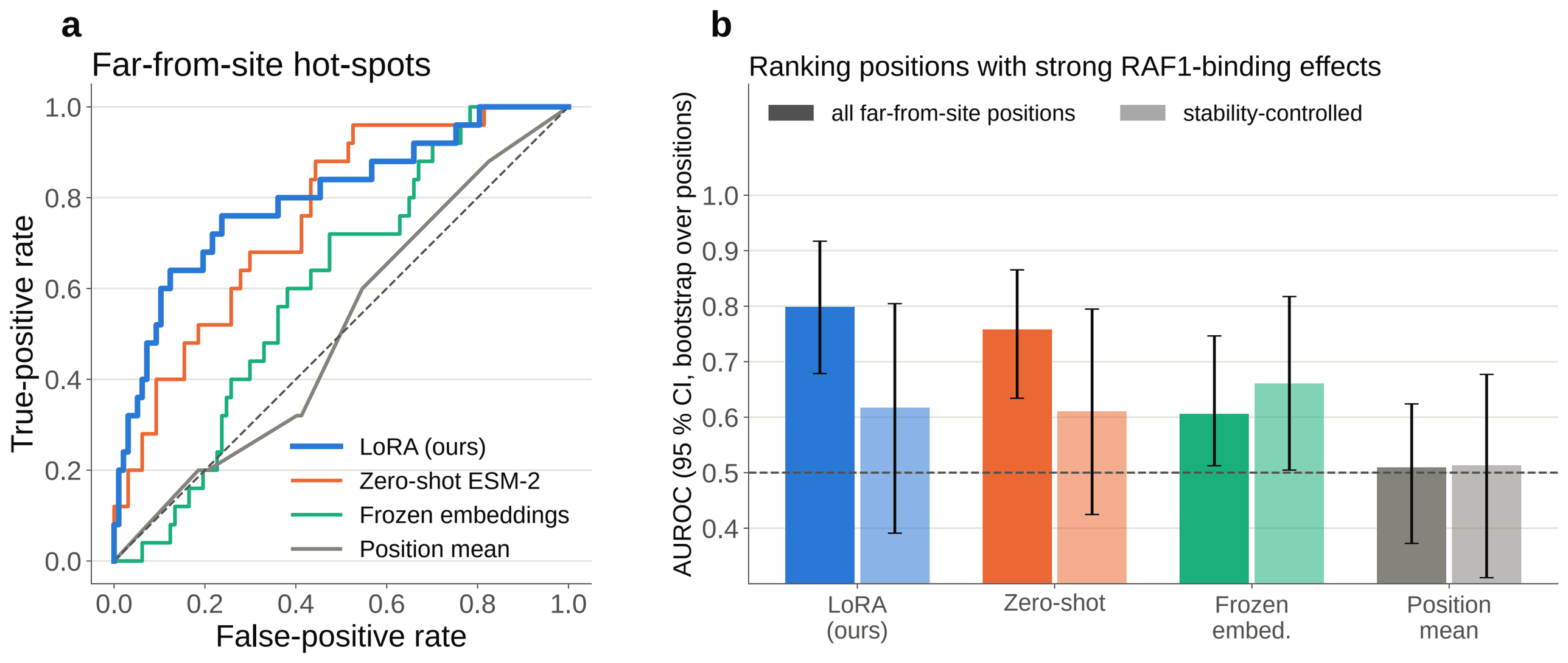
a
Far-from-site hot-spots
True-positive rate
False-positive rate
0.0
0.2
0.4
0.6
0.8
1.0
LoRA (ours)
Zero-shot ESM-2
Frozen embeddings
Position mean
b
Ranking positions with strong RAF1-binding effects
all far-from-site positions
stability-controlled
AUROC (95 % CI, bootstrap over positions)
0.4
0.5
0.6
0.7
0.8
0.9
1.0
LoRA (ours)
Zero-shot
Frozen embed.
Position mean